

itself. The Manufacturer MAY include a leaflet in the box with a larger QR code containing the concatenation of the four individual Onboarding Payloads. The user can then scan this combined QR code (one step for the user) which would give the Commissioner the Onboarding Payload for all four bulbs in one operation, and it can proceed to commission the four bulbs.

All Commissioners SHALL recognize the \* separator from the QR code as indication concatenation is used.

A Commissioner which does not support such concatenated Matter Onboarding Payloads SHOULD indicate to the user the need to commission devices one by one by scanning their individual QR codes or entering the individual Manual Pairing Codes.

The Commissioner SHOULD commission the devices in the order as they are provided in the concatenated code. (This ordering is particularly relevant in case of combi-packs where one of the devices needs to be commissioned first, e.g. a Thread Router first, then one or more Thread-connected bulbs).

Example of concatenated Onboarding Payloads:

```
MT:<onboarding-base-38-content_bulb1>*<onboarding-base-38-content_bulb2>*<onboarding-base-38-content_bulb3>*<onboarding-base-38-content_bulb4>
```

### 5.1.7. Generation of the Passcode

A device can support either dynamic or static passcodes for purposes of establishing the shared secret for the initial secure channel over which further commissioning steps take place.

All devices SHALL conform to the following rules for passcodes:

- Passcodes SHALL NOT be derived from public information, such as a serial number, manufacturer date, MAC address, region of origin, etc.
- The Passcode generation process SHALL use a cryptographically secure random number generator.

If a device generates a dynamic Passcode, then it SHALL conform to the following additional rule:

- Passcodes SHALL be accessible to commissioner only during the commissioning process.

If a device cannot generate a dynamic Passcode, then the static Passcode SHALL conform to the following additional rules:

- A random passcode with 27 bits of entropy SHALL be generated and used for each individual device. Because of the disallowed numbers, the entropy remaining in the actual passcode will be somewhere between 26 bits and 27 bits but the initial value before rejecting disallowed numbers SHALL have 27 bits of entropy.
- The device SHALL be supplied with the [PAKE](#) verifier in its internal storage.
- If the static passcode is also supplied to the device, the static passcode SHALL NOT be accessible during operational mode using any data model attributes or commands.
- If the static passcode is supplied to the device, its storage location SHALL be physically or logi-